Theory of Value

An Axiomatic Analysis of Economic Equilibrium
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Chapter 1: Mathematics

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02. Sets

03. Functions and Correspondences

04. Preorderings

05. Real Numbers

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08. Continuous Correspondences

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Introduction

This chapter presents all the mathematical concepts and results which will be used later (in notes, however, additional concepts and results will be freely used). Its reading requires, in principle, no knowledge of mathematics.

Introduction

The exposition starts from the concept of a set of elements and gradually introduces, by means of definitions, concepts of increasing complexity. Simultaneously results concerning those concepts are stated. Sections 1.2, 1.3, and 1.4 stay at a high level of generality; they deal respectively with sets, functions and correspondences, and preorderings. Section 1.5 is a crucial step in the exposition; it introduces (real) numbers. To prepare for their definition it is necessary to introduce earlier the concepts of a binary operation (in 1.3.j) and of a least upper hound (in 1.4.g). From the set $\mathbb{R}$ of (real) numbers, one builds up the Euclidean space of N dimensions, $\mathbb{R}^N$. Section 1.6 centers on the concept of a convergent sequence of points of $\mathbb{R}^N$. Sections 1.7 and 1.8 are respectively devoted to the concepts of continuity of functions and of correspondences. Section 1.9 is built on the definitions of the sum of two elements of $\mathbb{R}^N$ and of the product of an element of $\mathbb{R}^N$ by a (real) number; it considers $\mathbb{R}^N$ as a vector space. Section 1.10 studies the concept of fixed point of a correspondence.

Introduction

The method of exposition chosen parallels that of contemporary mathematics; one of its aims is to show mathematical concepts in their proper light. This is of invaluable help in the formulation and the solution of economic problems.

Introduction

It was stated that the reading of this chapter requires no knowledge of mathematics. This is, admittedly, true only "in principle." It certainly requires an ability to think abstractly, which is usually developed through the practice of mathematics, and an ability to assimilate in a short time a certain number of new concepts the motivation for which may not always be clear at first. On the other hand, the expert will notice that the logical foundations of set theory and even an elementary knowledge of the integers are taken for granted.

Introduction

The concepts and results which will be presented form the strict minimum necessary for a complete understanding of later chapters. For example, theorem (6) of I.6.n (every subset of $\mathbb{R}^N$ contains a dense countable set) is necessary for the proof of existence of a utility function in Chapter 4, section 6; theorem (16) of 1.9.x (the Minkowski bounding hyperplane theorem for convex subsets of $\mathbb{R}^N$) is, in essence, the central result on economic optimum which is proved in Chapter 6, section 4; theorem (2) of 1.10.d (the Kakutani fixed point theorem for correspondences from a convex subset of $\mathbb{R}^N$ to itself) is, in essence, the central result on economic equilibrium which is proved in Chapter 5, sections 6 and 7.

Introduction

Furthermore those concepts and results are almost all among the most basic of mathematics, and their usefulness goes far beyond the applications which are made of them below.

Introduction

Yet the reader may lack the time to read this entire chapter and the proofs of the economic theorems for which it prepares. The text of the following chapters has therefore been presented in such a way that the concepts and results stand out clearly and their meaning can be grasped with still less mathematical preparation.

Introduction

In this chapter, proofs of assertions are not given. In many cases the reader could reconstruct them; it might then be a valuable exercise for him to do so; hints will sometimes be provided. When this reconstruction offers difficulties the locution "One can prove..." or some explicit warning is used. Small-type passages contain examples and heuristic comments. They are irrelevant for the logical development of the text proper and could be omitted entirely; it is therefore permissible, in them, to draw upon an intuitive,knowledge of the physical world, and to use undefined simple mathematical terms like distance, curve, rectangle, ...